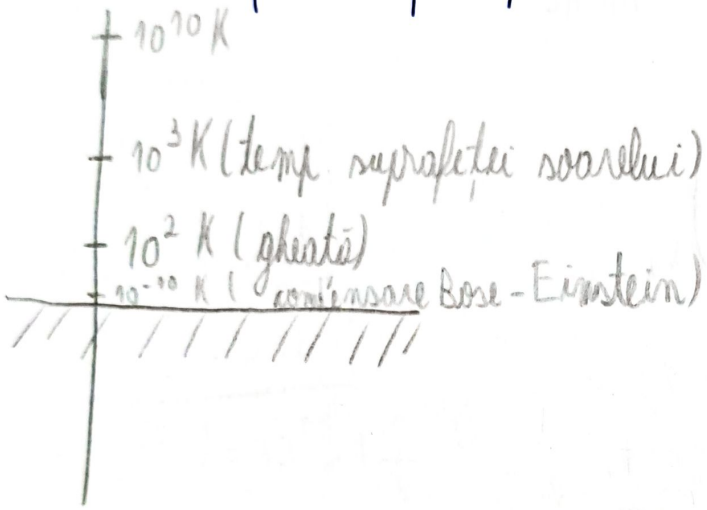
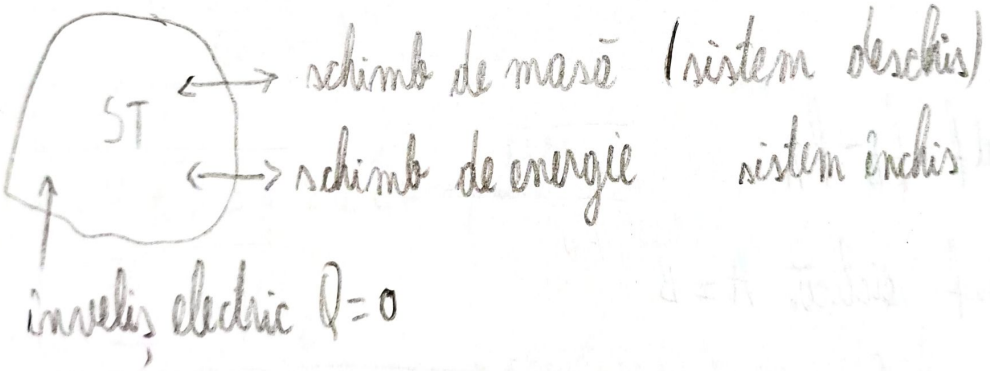
Fizică - curs 6

Noțiuni de termodinamică fizică

1. Mărimi fundamentale. Postulatul fundamental
2. Principiul 0. Tranzitivitatea echilibrului termic
3. Lucrul mecanic în termodinamică
4. Principiul 1:
   - 4.1. Capacități calorice și călduri latente
   - 4.2. Transformarea politropă





parametrii:

a) intensivi: p, T

b) extensivi ($V = V_1 + V_2$) prop de ~~aditivitate~~ aditivitate

$N = N_1 + N_2 \quad \vartheta = \vartheta_1 + \vartheta_2$

$$\vartheta = \frac{m}{\mu} = \frac{N}{N_A} = \frac{V}{V_\mu}$$

c) interni și externi

Mărimi [de stare → „uită" trecutul : U (energia internă) → „d"
[de proces → lucrul mecanic, Q [δL, δQ] → „δ" sau „∂"

↓

transformări ~~cvasi~~ cvasistatice

transformări reversibile – sistemul trece prin aceleași stări intermediare

↓

IDEAL



funcție de stare

$$df(x_1, x_2) = \left(\frac{\partial f}{\partial x_1}\right)_{x_2} + \left(\frac{\partial f}{\partial x_2}\right)_{x_1}$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = \frac{\partial^2 f}{\partial x_2 \partial x_1}$$

↓

$$\frac{\partial}{\partial x_1}\left(\frac{\partial f}{\partial x_2}\right) = \frac{\partial}{\partial x_2}\left(\frac{\partial f}{\partial x_1}\right)$$

d.t.e.

a) $\Delta f = \int_A^B df = f_B - f_A$

b) pt. transf. ciclică $A = B$

$\Delta f = 0$

c) $f(B) = \int_A^B df + const.$

$\frac{pV}{T} = ct.$

$pV = \nu RT$ (gaz ideal)

$\left(p + \frac{a}{V^2}\right)(V - b) = \nu RT$ (gazul Vandermas)

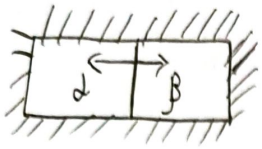
$R = R_0(1+\alpha)(t - t_0)$

$M = \frac{ct}{T}$ (magnetizație)

$l = l_0(1 + \alpha \Delta t)$

Postulatul fundamental: După un timp oarecare, se ajunge la echilibru termodinamic.

2.

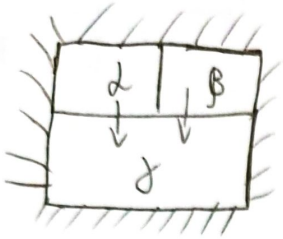


echilibru termic

$\alpha \overset{\text{ech. t.}}{\sim} \beta$ (reflexivitate)

$\beta \sim \alpha$

$\left.\begin{matrix} \alpha \sim \gamma \\ \beta \sim \gamma \end{matrix}\right| \Rightarrow \alpha \sim \beta$ (tranzitivitate)



Scara Celsius

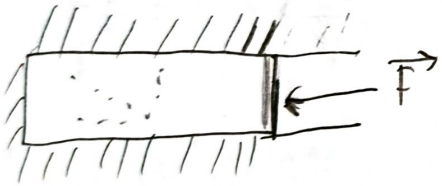
0°C ... 100°C

Scara Fahrenheit

32°C ... 212°F

$t(^\circ F) = 32^\circ C + \frac{9}{5}\,^\circ C$

3.



$\delta L = F\,dl = p\,S\,dl = -p\,dV$

$dL = \sum_{i=1}^{n} Y_i\,dx_i$ → coordonate generalizate

↓ forțe generalizate

alte ex: $\delta L = -\vec{M}\,d\vec{B}$

↳ magnetație

$\delta L = \sigma \, dS$

↓ forță de tensiune superficială

$\delta L = -\vec{P} \, d\vec{e}$

↓ polarizație

$\delta L = -\vec{H} \, d\vec{B}$ etc.

pt. gaz ideal

a) proces izocor $V = ct \Rightarrow dV = 0$

$\delta L = 0 \Rightarrow L = 0$

b) proces izobar: $p = ct$

$$L_{AB} = \int_A^B \delta L = \int_A^B p \, dV = p \int_A^B dV = p(V_B - V_A)$$

c) proces izoterm: $T = ct.$

$$L_{AB} = \int_A^B p \, dV = \nu RT \int_A^B \frac{dV}{V} = \nu RT \ln \frac{V_B}{V_A}$$

↓

$$p = \frac{\nu RT}{V}$$

$\boxed{\Delta U = Q - L}$ Legea de conservare a energiei

Cazuri particulare

a) pt. proces izocor: $L = 0 \Rightarrow \Delta U = Q$ $Q > 0$ (căldură acceptată)

$Q < 0$ (— " — cedată mediului)

b) pt. transf. ciclice: $\Delta U = 0$

$Q = L$

c) pt. transf. adiabată: $Q = 0$

$\Delta U = L$

$$\begin{cases} \lambda_\mu = \left(\frac{\partial H}{\partial \mu}\right)_T - V \\ C_\mu = \left(\frac{\partial H}{\partial T}\right)_\mu \end{cases} \quad \begin{cases} C - C_\mu = \lambda_\mu \frac{d\mu}{dT} \\ C - C_V = \lambda_V \frac{dV}{dT} \end{cases}$$

pt. gaz ideal: $U = U(t) \Rightarrow \left(\frac{\partial U}{\partial V}\right)_T = 0$

$$H = H(t) \Rightarrow \left(\frac{\partial H}{\partial \mu}\right)_T = 0 \Rightarrow \lambda_\mu = -V$$

$$\frac{C - C_\mu}{C - C_V} = -\frac{V}{\mu} \frac{d\mu}{dt} \cdot \frac{dt}{dV}$$

n = indicele transf. $\not\Rightarrow$ politrop

$C = ct.$ $\qquad \mu V^n = ct.$

$n = -\frac{V}{\mu}\frac{d\mu}{dV}$ $\qquad \mu V = \nu RT$

$\int \frac{d\mu}{\mu} = -n \int \frac{dV}{V}$ $\qquad \mu \sim \frac{T}{V} \Rightarrow TV^{n-1} = ct.$

Cazuri particulare $\qquad \boxed{n = \frac{C - C_\mu}{C - C_V}}$

a) $n = 0$: $C = C_\mu$ (transf. izobară) $\Rightarrow \frac{V}{T} = ct.$

b) $n = 1$: $C \to \infty$: $n = \frac{1 - \frac{C_\mu}{C}}{1 - \frac{C_V}{C}} \to 1$

$\Downarrow$

cf. def. $dT \to 0 \Rightarrow T = const. \Rightarrow$ tr. izotermă $\mu V = ct.$

c) $n = \gamma = \frac{C_\mu}{C_V} \Rightarrow C = 0 \Rightarrow$ și $Q = 0 \Rightarrow$ tr. adiabatică $\Rightarrow \mu V^\gamma = ct.$

d) $n \to \infty$: $C = C_V \Rightarrow$ tr. izocoră $\frac{\mu}{T} = ct.$

4.1. $C = \frac{\delta Q}{dT}$ capacitatea calorică (căldura schimbată pt. modif. temp. cu 1K)

$C_p = \left(\frac{\partial Q}{\partial T}\right)_p \quad C_V = \left(\frac{\partial Q}{\partial T}\right)_V$

$\left.\begin{aligned} \lambda_p &= \left(\frac{\partial Q}{\partial p}\right)_T \\ \lambda_V &= \left(\frac{\partial Q}{\partial V}\right)_T \end{aligned}\right\}$ călduri latente

$\boxed{Q = Q(V,T)}$

$\delta Q = \left(\frac{\partial Q}{\partial V}\right)_T dV + \left(\frac{\partial Q}{\partial T}\right)_V dT \;/:dT$

$C = \lambda_V \frac{dV}{dT} + C_V$

$\delta Q = dU + p\,dV = \left(\left(\frac{\partial U}{\partial V}\right)_T + p\right) dV + \left(\frac{\partial U}{\partial T}\right)_V dT$

$\Rightarrow \left.\begin{aligned} \lambda_V &= p + \left(\frac{\partial U}{\partial V}\right)_T \\ C_V &= \left(\frac{\partial U}{\partial T}\right)_V \end{aligned}\right\}$

$\boxed{Q = Q(p,T)}$

$\delta Q = \left(\frac{\partial Q}{\partial p}\right)_T dp + \left(\frac{\partial Q}{\partial T}\right)_p dT$

$C = \lambda_p \frac{dp}{dT} + C_p$

entalpia: $H = U + pV \Rightarrow dH = dU + p\,dV + V\,dp = \delta Q + V\,dp$

$\delta Q = dH - V\,dp = \left[\left(\frac{\partial H}{\partial p}\right)_T - V\right] dp + \left(\frac{\partial H}{\partial T}\right) dT$

Curs 7

1. Principiul al 2-lea al termodinamicii
1.1. Formulări
1.2. Ciclul Carnot. Teoremele lui Carnot
2. Entropie. Legea creșterii entropiei
3. Entropia gazului ideal
4. Interpretarea microscopică a entropiei
5. Funcții caracteristice. Potențiale termodinamice (U, F, G, H)
6. Criterii de echilibru pt. ST
7. Fenomenul magnetocaloric. Aplicații. Răcirea adiabatică prin proces de magnetizare)
8. Principiul al 3-lea

① P1: $dU = \delta Q - p\,dV$

$\Delta U = Q - L$

$\boxed{\Delta U = 0} \Rightarrow Q = L$ transf. ciclică

↓

a) conducție ($j_Q = -K_T \cdot \text{grad}\, T$)

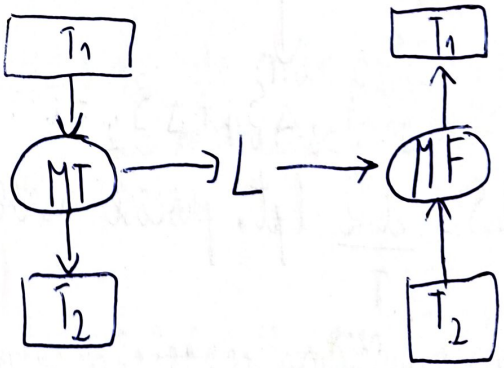
b) radiație ($\sim \sigma T^4$)

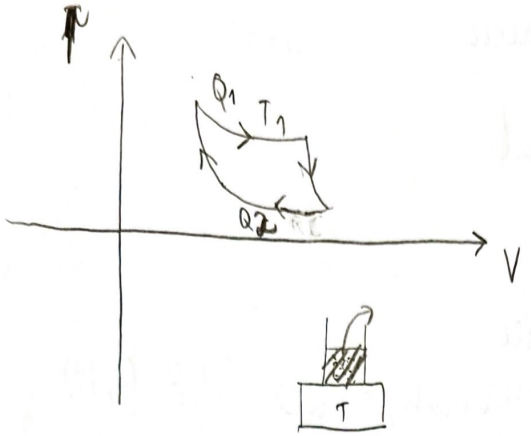
c) convecție

formulări

① Thompson

②





$$\eta_C = \frac{L}{Q_1} = 1 - \frac{|Q_2|}{Q_1}$$

$$\eta_C = 1 - \frac{T_2}{T_1}$$

(T1) $\eta = f(T_1, T_2)$ nu depinde de natura substanței

$T_2 = 273{,}15\,K = T_0 \Rightarrow T$ temp. absolută

η_C reversibil $> \eta_C$ ireversibil (cel mai mare randament posibil este cel al ciclului Carnot) → (condensare a căldurii) → căldură de compensare

(2) $1 - \frac{|Q_2|}{Q_1} = 1 - \frac{T_2}{T_1} \Rightarrow \frac{Q_1}{T_1} = \frac{|Q_2|}{T_2}$

$\Rightarrow \boxed{\frac{Q_1}{T_1} + \frac{Q_2}{T_2} = 0}$

căldură redusă

$\boxed{\Delta S = \frac{Q}{T}}$ entropia (căldura schimbată la temp. const.)

$\Delta S_1 + \Delta S_2 = 0$

$dS = \frac{\delta Q}{T}$ (pt. procese reversibile)

$\sum_{i=1} \frac{Q_i}{T_i} \overset{nu}{=} 0$ (procese reversibile)

Prop.

a) funcție de stare

b) $\Delta S = \int_A^B \frac{đQ}{T}$

c) $S_B = S_A + \int_A^B \frac{đQ}{T}$

$\oint \frac{dQ}{T} \underset{rev.}{\overset{irev.}{\leq}} 0 \longrightarrow \Delta S \underset{rev.}{\overset{irev.}{\geq}} \oint \frac{đQ}{T}$ pt. procese izolate

Toate procesele din natură se desfășoară în sensul creșterii entropiei.

③ $dU = T\,dS - p\,dV$

$dS = \frac{1}{T} dU + \frac{p}{T} dV$

$C_V = \left(\frac{\partial Q}{\partial T}\right)_V - \left(\frac{\partial U}{\partial T}\right)_V \rightarrow \frac{\partial R}{V}$

$\int_A^B dS = \int_A^B \frac{C_V}{T} dT + \int_A^B \left(\frac{p}{T}\right) dV$

$S_B - S_A = C_V \ln \frac{T_B}{T_A} - \partial R \ln \frac{V_B}{V_A}$

④ $\boxed{S = K_B \ln \Omega}$

Ω → probabilitate termodinamică (nr. de microstări compatibile cu o microstare dată)

K_B → const. Boltzmann

entropia informațională (Shannon):

$Q = -K \ln P$ P = probabilitatea de realizare

$1\mu \to 2$

$N_A \to 2N$

$S = k_B \ln 2^N = N k_B \ln 2$ entropia unui sistem de N_A particule cu 2 stări

$S = R \ln 2$

$Q^{inf} = \langle Q^{inf} \rangle = \sum_{(i)} Q_i P_i = -K \sum_{(i)} P_i \ln P_i$ dacă sunt mai multe evenimente urmărite simultan

$\Delta S_{univ} = \Delta S_{sist} + \Delta S_{ME}$ → crește

↓ scade

⑤ T, V, μ, S

a) energie internă:

$dU = T\,dS - \mu\,dV$

$U = U(S, V)$

$dU = \left(\frac{\partial U}{\partial S}\right)_V dS + \left(\frac{\partial U}{\partial V}\right)_S dV$

$T = \left(\frac{\partial U}{\partial S}\right)_V ; \left(\frac{\partial}{\partial V}\right)_S$

$\mu = -\left(\frac{\partial U}{\partial V}\right)_S ; \left(\frac{\partial}{\partial S}\right)_V$

$\Rightarrow \frac{\partial^2 U}{\partial S \partial V} = \frac{\partial^2 U}{\partial V \partial S} \Rightarrow \boxed{\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial \mu}{\partial S}\right)_V}$ relațiile Maxwell

$\Delta U = \int (dU)_V = \int (T\,dS)_V = \int C_V\,dT$

b) entalpia: $\textcircled{H} = H(S, p)$

$H = U + pV$

$dH = T\,dS + V\,dp$

$T = \left(\frac{\partial H}{\partial S}\right)_p \quad \Big| \frac{\partial}{\partial p}$

$V = \left(\frac{\partial H}{\partial p}\right)_S \quad \Big| \frac{\partial}{\partial S}$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$$

c) energia liberă: $\boxed{F = F(V, T)}$ (fct. Helmholtz)

$F \overset{def}{=} U - TS$

$dF = -p\,dV - S\,dT$

$p = -\left(\frac{\partial F}{\partial V}\right)_T \quad \Big| \frac{\partial}{\partial T} \quad \Rightarrow \left(\frac{\partial p}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T$

$S = -\left(\frac{\partial F}{\partial T}\right)_V \quad \Big| \frac{\partial}{\partial V}$

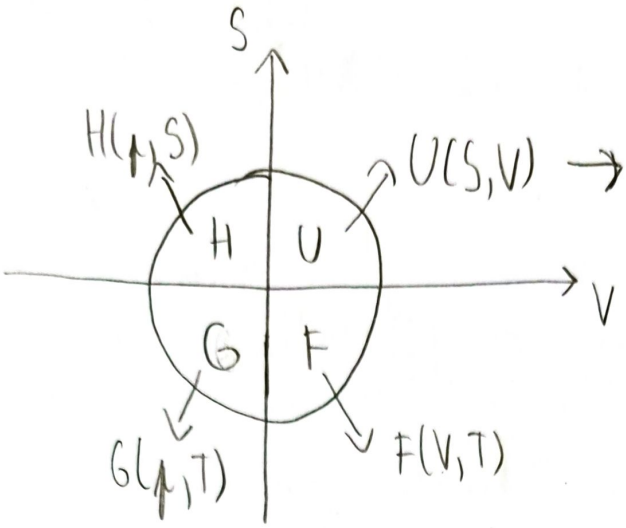
d) entalpie liberă (fct. GIBBS) $G = G(p, T)$

$G = H - TS$

$dG = V\,dp - S\,dT$

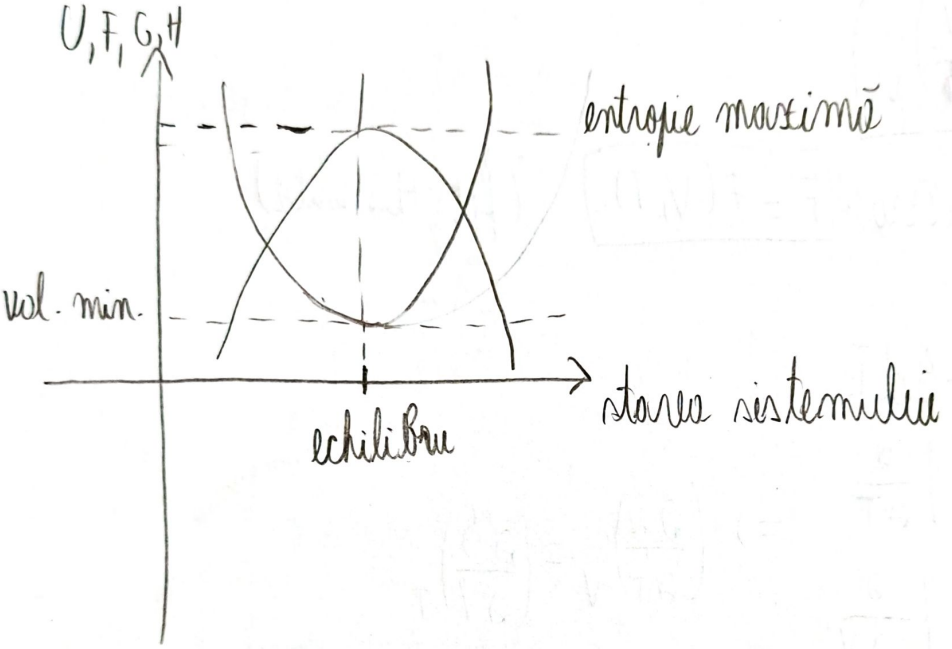
$V = \left(\frac{\partial G}{\partial p}\right)_T \quad \Rightarrow \boxed{\left(\frac{\partial V}{\partial T}\right)_p = -\left(\frac{\partial S}{\partial p}\right)_T}$

$S = -\left(\frac{\partial G}{\partial T}\right)_p$



$dU = T\,dS - p\,dV$

⑥



⑦ (ST) → substanță magnetică

$\vec{H}$; $dL = \vec{H}\,d\vec{B}$; V = 1

intensitatea câmpului magnetic

$\vec{B} = \mu_0 \mu_{rel} \vec{H} = \mu_0 (1 + \chi_n) \vec{H}$

„χ"
hi

susceptibilitate magnetică χ_n ↓ $\frac{\chi}{H}$ magnetizație

const.

$\chi_n = \frac{C}{T}$ legea Curie

$$dG = -S\,dT + B\,dH$$
$$= -S\,dT + \mu_0(1+\chi_m)\cdot H\,dH \;;\; G = G(T,H)$$

$$S = -\left(\frac{\partial G}{\partial T}\right)_H$$

$$\mu_0(1+\chi_m)\,H\,dH = \left(\frac{\partial G}{\partial H}\right)_T \Rightarrow \int_0^H \left(\frac{\partial G}{\partial H}\right)_T dH = \int_0^H \mu_0(1+\chi_m)\,H\,dH$$

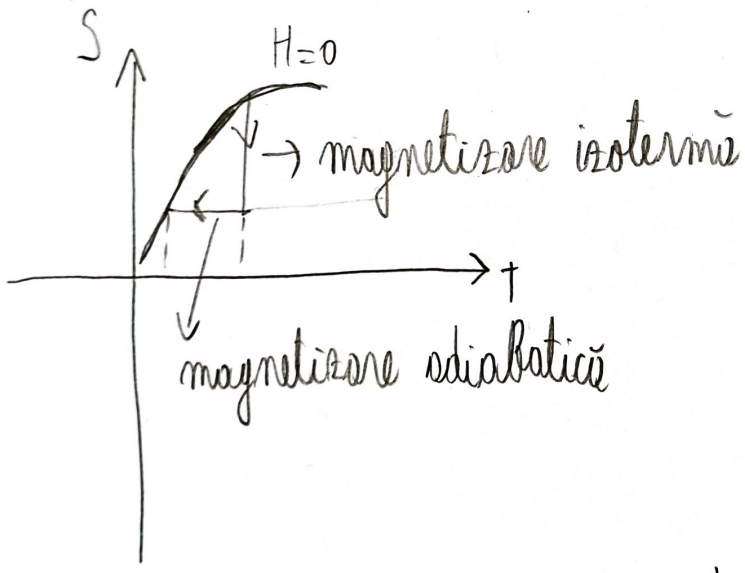
$$G(H,T) - G(0,T) = \mu_0(1+\chi_m)\frac{H^2}{2}$$

$$\Downarrow$$

$$S(H,T) - S(0,T) = -\mu_0 \left(\frac{d\chi_m}{dT}\right)\frac{H^2}{2}$$

$$\downarrow$$

$$-\frac{C}{T^2}$$



(8) (P3) imposibilitatea stingerii lui 0K